

Target validation paths — mcrc-kras-a59t-pik3ca-peritoneal-pre-hi pec-q8k4

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

Seven essential gates and four high-priority workups have to land before the rest of the ranking moves from biomarker-conditional to actionable: comprehensive tissue NGS (FoundationOne CDx or equivalent) returns KRAS A59T variant allele fraction plus full extended-RAS clearance, BRAF V600, ERBB2 amplification, PIK3CA M1043I clonality, TP53 R273 substitution, SMAD4, and the 20q segment boundary; HER2 IHC/FISH per HERACLES is the most-actionable target check not yet done and gates trastuzumab deruxtecan via NCT04744831 (RAS-mutant-allowed) plus the RAS-WT-restricted tucatinib + trastuzumab door via MOUNTAINEER-03 (NCT05253651); MMR IHC orthogonal to NGS-MSI closes the ICI door definitively rather than presumptively; TMB in mut/Mb with the panel definition documented locks the tumor-agnostic pembrolizumab gate; PD-L1 IHC with clone + CPS covers combination-trial stratification; DPYD and UGT1A1 germline genotyping land the March 2024 FDA fluoropyrimidine label update and the 2026 NCCN recommendation; Signatera tumor-informed baseline before CRS gives the lead-time signal for peritoneal recurrence after surgery. None of this requires a new biopsy. All seven essential gates can run in parallel off the archival FFPE block plus a peripheral blood draw plus a pre-CRS Signatera tube.

KRAS A59T

Essential: comprehensive tissue NGS (FoundationOne CDx, Tempus xT CDx, Caris MI Cancer Seek, or MSK-IMPACT) with explicit KRAS A59T variant allele fraction, full extended RAS coverage (KRAS / NRAS exons 2/3/4, HRAS), and an explicit BRAF V600 call. A59 substitutions are recognized RAS-pathway activating alterations and current NCCN consensus treats them as resistance variants for anti-EGFR therapy, but Lochhead 2018 reported a -36% target-lesion change on panitumumab + FOLFIRI in a single A59T tumor and Arena 2021 characterized A59 alleles as showing weaker MAPK signaling than canonical G12 mutants. A clean VAF, a fully resolved extended RAS, and a documented BRAF V600 status are the only way to scope cetuximab / panitumumab and BEACON (encorafenib + cetuximab) eligibility for this patient. The same panel returns ERBB2 amplification, MSI by NGS, TMB in mut/Mb, and the PIK3CA / APC / TP53 / SMAD4 confirmations every downstream row depends on. Turnaround 2-3 weeks; archival FFPE from the resected liver metastasis or the primary is sufficient.

Essential: HER2 IHC (clone 4B5 or HercepTest) with reflex dual-probe FISH per MOUNTAINEER / HERACLES scoring (IHC 3+ in >50% of cells, or IHC 2+ with HER2:CEP17 ratio ≥ 2.0). HER2 amplification covers another 3-5% of mCRC and is the single most-actionable target check not yet done on this case. A positive return routes trastuzumab deruxtecan via DESTINY-CRC02 (NCT04744831, ORR ~38% including RAS-mutant under the 2024 pan-tumor accelerated approval); the MOUNTAINEER tucatinib + trastuzumab door is RAS-WT-restricted by label and the A59T patient is closed out regardless of HER2 status. Order the IHC first; FISH only runs if the IHC reads 2+ equivocal. Turnaround 5-10 business days.

Low priority: documentation review of the existing negative multi-gene germline panel to confirm POLE / POLD1 proofreading-domain coverage (POLE P286R / V411L hotspots). A 37-year-old sits in the early-onset bracket where polymerase-proofreading deficiency is on the differential even at TMB-low; if the prior panel did not specifically cover the proofreading exonuclease domain, a focused add-on is warranted. Otherwise this is a chart-review row.

TP53 R273

Essential: explicit TP53 R273 substitution call (R273H vs R273C vs R273L vs R273S vs R273P) from the comprehensive NGS report. R273H and R273C have the strongest preclinical APR-246 / eprenetapopt and arsenic-trioxide reactivation data, R273L is intermediate, and R273S is a less-studied DNA-contact variant. The eprenetapopt solid-tumor program is effectively closed after Park 2022 posted 2 PRs / 24 evaluable, and PYNACLE rezatapopt is Y220C-binding-pocket-restricted and additionally KRAS-WT-only by protocol so this patient is structurally ineligible regardless of the R273 sub-call. Resolving the substitution is still load-bearing because it sharpens any future ATR / WEE1 / HSP90 synthetic-lethal trial discussion at recurrence. Comes off the master NGS report; the action is to request the lab to call the substitution explicitly rather than abbreviating to a generic 'p.R273'.

Low priority: confirmation that no co-occurring TP53 Y220C variant is present on the existing NGS report. Most TP53-mutant CRC tumors carry a single dominant alteration; PYNACLE requires Y220C plus KRAS wild-type, and the case fails the KRAS-WT gate regardless, but the rule-out is documented so the trial screener does not waste time on it.

MSS / TMB-low / PD-L1-negative

Essential: MMR protein IHC four-antibody panel (MLH1, MSH2, MSH6, PMS2) on archival FFPE, paired with the NGS-derived MSIsensor or MANTIS score from the master panel. MSS by a single method misses the rare but real discordant case where NGS-MSI or IHC alone calls a tumor MSS that the orthogonal method calls MSI-H (concordance ~95%, discordance ~5%). Confirming MSS by both methods closes the ICI door definitively rather than presumptively, which matters because MSI-H mCRC has a 44% ORR with pembrolizumab and would reorder the entire decision tree. Turnaround 5-7 business days; the IHC is a low-cost defensive call that the comprehensive NGS does not substitute for.

Essential: explicit TMB value in mutations per megabase from the comprehensive NGS report, with the panel definition and threshold documented. The intake says 'TMB-low' without the numeric value, and 8-9 mut/Mb closes the door while 10-12 opens the tumor-agnostic pembrolizumab label. Each panel uses a slightly different bait region and counting rule (FoundationOne CDx 0.8 Mb of CDS; Tempus xT 2.4 Mb reportable; MSK-IMPACT 1.06 Mb), so the panel name and threshold travel with the number. Bundled with the master NGS.

Essential: PD-L1 IHC with explicit clone (22C3, SP263, or 28-8) and explicit scoring system (CPS for mCRC). 'Negative' without a clone or scoring system is not interpretable for trial eligibility because different ICI labels and combination studies use different cutoffs and antibodies. Several MSS-mCRC combination ICI trials use PD-L1 CPS thresholds for stratification rather than for hard exclusion, so the exact CPS value still matters even when the headline result is negative. Default to 22C3 with CPS scoring for CRC context.

Medium priority: high-resolution HLA-A typing (4-digit) plus full class-I and class-II profile on peripheral blood. MSS / TMB-low mCRC is the population where HLA-restricted TCR-T (afami-cel, lete-cel, IMA203) and ImmTAC (brenetafusp) trials are most often considered; HLA-A*02:01 is the dominant restriction. Bank the result before the HIPEC procedure even if no TCR-T trial is in immediate play; the result lasts for life and removes friction from any future referral.

Peritoneal carcinomatosis

Essential: DPYD genotyping for the four CPIC-recommended variants (c.1905+1G>A / 2A, c.1679T>G / 13, c.2846A>T, c.1236G>A / HapB3) on peripheral whole blood. The March 2024 FDA fluoropyrimidine label update and the 2026 NCCN colon guideline both advise pre-treatment testing. The patient is already on FOLFIRI and has

tolerated cycles, which lowers but does not eliminate the value: heterozygous carriers of c.2846A>T or HapB3 can show progressive cumulative toxicity, and the HIPEC-window dose decision plus any future capecitabine maintenance both benefit from a documented genotype.

High priority: UGT1A1 6 and 28 genotyping bundled with DPYD on a single pharmacogenomic order. 28/28 (and 6/28 in Asian ancestry) confers a 3-4x increased risk of severe irinotecan-related neutropenia and diarrhea, and the FDA irinotecan label recommends a starting-dose reduction in homozygotes. Pre-resolves the irinotecan-HIPEC question if that perfusate route enters discussion.

High priority: Signatera tumor-informed serial ctDNA MRD at three peri-HIPEC timepoints (pre-CRS baseline, post-operative day 30-60, then every 8-12 weeks). Peritoneal-spread CRC is one of the settings where ctDNA MRD is most predictive: pre-CRS ctDNA detection correlates with shorter PFS and identifies patients who may benefit more from extended systemic therapy than from upfront cytoreduction. Signatera carries Medicare coverage for stage II-IV CRC MRD surveillance and the tumor-informed bespoke design (16 patient-specific variants from WES) gives the best low-tumor-fraction sensitivity; Guardant Reveal is the tumor-naïve alternative if faster start outweighs sensitivity. The bespoke build runs 4-6 weeks, so the order has to land BEFORE CRS for the baseline draw to be interpretable.

Medium priority: serum CEA, CA 19-9, and CA 125 at baseline and at each peri-HIPEC interval. CA 125 elevation correlates with peritoneal carcinomatosis burden and tracks response and recurrence independently of CEA. Together they give a low-cost peri-HIPEC response signal that complements ctDNA and imaging. Pull all three at every routine draw rather than CEA alone; the marginal cost is negligible.

BCL2L1 / TOP1 (20q11)

High priority: 20q amplification segment definition (20q11.21 BCL2L1 / TOP1 vs extended 20q13 ZNF217 / MYBL2 / AURKA) from the existing NGS copy-number plot, with optional confirmatory FISH or OncoScan if the NGS boundaries are ambiguous. 20q11.21-restricted amplification centered on BCL2L1 (BCL-xL) and TOP1 is biologically distinct from broad 20q gain: the BCL2L1 subset supports BH3-mimetic / BCL-xL-degrader rationale (navitoclax solid-tumor program discontinued; DT2216 pediatric-only trial) and the TOP1 amplification correlates with irinotecan sensitivity, while the broader 20q gain raises AURKA-class trial questions instead. No new sample needed in most cases; the action is on interpreting the existing CNV plot. If the report flags 20q amplification without segment boundaries, request a CNV-segment annotation from the lab or reflex to OncoScan.

SMAD4 R361H

High priority: consensus molecular subtype (CMS1-4) by transcriptomic classifier (CRCAssigner / CMScaller from RNA-seq) plus CDX2 IHC and the ZEB1 / HTR2B / FRMD6 / pan-cytokeratin IHC surrogate panel when RNA-seq is not available. Peritoneal-dominant disease with SMAD4 loss raises the prior for CMS4 (mesenchymal) classification, which has a worse prognosis but identifies patients with a TGF-beta-mediated immune-excluded phenotype and a different combination-therapy landscape than CMS2 / CMS3. The IHC surrogate panel achieves 87% concordance with transcriptomic CMS and is the practical clinical-lab path when full RNA-seq is unavailable. Best value if Caris MI Cancer Seek is already used as the master comprehensive panel since CMS comes off the whole-transcriptome data.

Medium priority: SMAD4 (DPC4) IHC (clone B-8 or EP618Y) on archival FFPE. R361H sits in the MH2 DNA-binding domain and is a known loss-of-function hotspot; SMAD4 IHC loss is the protein-level orthogonal confirmation that the molecular call translates to absent nuclear SMAD4. Cheap defensive call rather than a gating

one; pair on a single block release with the MMR IHC.

Low priority: spatial multiplex immune profile of TIL density, CD3 / CD8 distribution, FAP+ stromal CAF density, and pSMAD2/3 in the peritumoral stroma (Akoya PhenoCycler / Lunaphore platform). SMAD4 loss plus a CMS4 phenotype is the canonical TGF-beta-driven immune-excluded state where CD8+ T cells are stuck at the invasive margin behind a FAP+ CAF stromal collar. A multiplex profile separates 'cold' from 'excluded' cases and sharpens any anti-TGF-beta + ICI combination-trial discussion. Research-grade; reserve for the moment a TGF-beta-axis or stromal-targeting protocol enters serious discussion.

APC E1295

Medium priority: exact APC codon-1295 variant type (E1295* nonsense vs E1295fs frameshift vs missense) and position relative to the mutation cluster region (MCR, codons 1250-1500), with second-allele status (LOH vs second truncating event) from the existing NGS report. Codon 1295 sits within the MCR; truncating events in the MCR retain partial beta-catenin binding but lose the SAMP repeats needed for downregulation, and bi-allelic MCR truncation is the typical 'just right' Wnt-pathway state most studied for Wnt-axis drug rationale. The variant call and second-allele status decide whether this case is a strong candidate for Wnt-pathway investigational trials (PORCN inhibitors, tankyrase inhibitors, frizzled antibodies) or a weak one. No new sample; request the lab to explicitly call the variant type and the second-allele state, both of which sit in the raw VCF but are sometimes summarized.

Where to order these assays

Assay	Provider	Decision gated	Contact
Comprehensive tissue NGS (KRAS A59T VAF + extended RAS + BRAF V600 + ERBB2 + PIK3CA + TP53 R273 + SMAD4 + 20q + TMB)	Foundation Medicine (preferred) (FoundationOne CDx)	Anti-EGFR (cetuximab, panitumumab) and BEACON (encorafenib + cetuximab) eligibility; resolves the dominant first-question gate on systemic-therapy options.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Comprehensive tissue NGS	Tempus Labs (Tempus xT CDx)	Anti-EGFR (cetuximab, panitumumab) and BEACON (encorafenib + cetuximab) eligibility; resolves the dominant first-question gate on systemic-therapy options.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Comprehensive tissue NGS	Caris Life Sciences (MI Cancer Seek)	Anti-EGFR (cetuximab, panitumumab) and BEACON (encorafenib + cetuximab) eligibility; resolves the dominant first-question gate on systemic-therapy options.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
Comprehensive tissue NGS	NeoGenomics Laboratories (NeoTYPE Comprehensive Tumor Profile)	Anti-EGFR (cetuximab, panitumumab) and BEACON (encorafenib + cetuximab) eligibility; resolves the dominant first-question gate on systemic-therapy options.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907

Comprehensive tissue NGS	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	Anti-EGFR (cetuximab, panitumumab) and BEACON (encorafenib + cetuximab) eligibility; resolves the dominant first-question gate on systemic-therapy options.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
HER2 (ERBB2) IHC clone 4B5 / HercepTest with reflex dual-probe FISH per HERACLES	NeoGenomics Laboratories (preferred) (HER2 IHC + FISH (HERACLES/MOUNTAINEER scoring available))	Tucatinib + trastuzumab (MOUNTAINEER label, RAS-WT only), trastuzumab deruxtecan off-label, HERACLES regimens, and MOUNTAINEER-03 first-line study (NCT05253651).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
HER2 IHC + reflex FISH	Mayo Clinic Laboratories (HER2 IHC + Dual-Probe FISH)	Tucatinib + trastuzumab (MOUNTAINEER label, RAS-WT only), trastuzumab deruxtecan off-label, HERACLES regimens, and MOUNTAINEER-03 first-line study (NCT05253651).	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
HER2 IHC + reflex FISH	Labcorp (HER2/ERBB2 IHC with Reflex FISH)	Tucatinib + trastuzumab (MOUNTAINEER label, RAS-WT only), trastuzumab deruxtecan off-label, HERACLES regimens, and MOUNTAINEER-03 first-line study (NCT05253651).	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
HER2 IHC + reflex FISH	Quest Diagnostics (HER2 IHC + FISH)	Tucatinib + trastuzumab (MOUNTAINEER label, RAS-WT only), trastuzumab deruxtecan off-label, HERACLES regimens, and MOUNTAINEER-03 first-line study (NCT05253651).	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
MMR protein IHC four-antibody panel (MLH1, MSH2, MSH6, PMS2) plus NGS-MSI orthogonal	NeoGenomics Laboratories (preferred) (MMR IHC Panel (MLH1/MSH2/MSH6/PMS2))	Single-agent and dual-agent ICI (pembrolizumab, dostarlimab, nivolumab + ipilimumab); reverses entire treatment paradigm if MSI-H by either method.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
MMR protein IHC	Mayo Clinic Laboratories (MMR Protein IHC)	Single-agent and dual-agent ICI (pembrolizumab, dostarlimab, nivolumab + ipilimumab); reverses entire treatment paradigm if MSI-H by either method.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
MMR protein IHC	Labcorp (MMR Protein IHC)	Single-agent and dual-agent ICI (pembrolizumab, dostarlimab, nivolumab + ipilimumab); reverses entire treatment paradigm if MSI-H by either method.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167

MMR protein IHC	Quest Diagnostics (MMR Protein IHC Panel)	Single-agent and dual-agent ICI (pembrolizumab, dostarlimab, nivolumab + ipilimumab); reverses entire treatment paradigm if MSI-H by either method.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
TMB explicit mut/Mb value with panel + threshold documented	Foundation Medicine (preferred) (FoundationOne CDx (FDA-labeled TMB-H companion diagnostic))	Tumor-agnostic pembrolizumab eligibility (TMB-H ≥10 mut/Mb label).	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
TMB explicit mut/Mb	Tempus Labs (Tempus xT CDx)	Tumor-agnostic pembrolizumab eligibility (TMB-H ≥10 mut/Mb label).	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
TMB explicit mut/Mb	Caris Life Sciences (MI Cancer Seek)	Tumor-agnostic pembrolizumab eligibility (TMB-H ≥10 mut/Mb label).	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
PD-L1 IHC with explicit clone (22C3 / SP263 / 28-8) and explicit scoring (CPS for CRC)	NeoGenomics Laboratories (preferred) (PD-L1 22C3 pharmDx (CPS) and SP263 (TPS))	ICI combination trial stratification and any PD-L1-conditioned MSS-CRC ICI regimen.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
PD-L1 IHC	Mayo Clinic Laboratories (PD-L1 22C3 / SP263 / 28-8 IHC)	ICI combination trial stratification and any PD-L1-conditioned MSS-CRC ICI regimen.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
PD-L1 IHC	Labcorp (PD-L1 22C3 IHC)	ICI combination trial stratification and any PD-L1-conditioned MSS-CRC ICI regimen.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
PD-L1 IHC	Quest Diagnostics (PD-L1 IHC)	ICI combination trial stratification and any PD-L1-conditioned MSS-CRC ICI regimen.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
DPYD four-variant CPIC genotyping (2A, 13, c.2846A>T, HapB3)	Mayo Clinic Laboratories (preferred) (DPYD Genotype (test code DPYDZ))	Fluoropyrimidine dose adjustment for ongoing FOLFIRI and any future capecitabine maintenance.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
DPYD genotyping	Labcorp (DPYD Genotyping)	Fluoropyrimidine dose adjustment for ongoing FOLFIRI and any future capecitabine maintenance.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
DPYD genotyping	Quest Diagnostics (DPYD Genotype)	Fluoropyrimidine dose adjustment for ongoing FOLFIRI and any future capecitabine maintenance.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
DPYD genotyping	ARUP Laboratories (DPYD Genotyping, 5 Variants)	Fluoropyrimidine dose adjustment for ongoing FOLFIRI and any future capecitabine maintenance.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-522-2787
UGT1A1 6 / 28 genotyping (TA repeat polymorphism)	Mayo Clinic Laboratories (preferred) (UGT1A1 Genotyping (test code UGTGT))	Irinotecan dose adjustment in FOLFIRI and any irinotecan-based HIPEC protocol.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
UGT1A1 genotyping	Labcorp (UGT1A1 Genotype)	Irinotecan dose adjustment in FOLFIRI and any irinotecan-based HIPEC protocol.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167

UGT1A1 genotyping	Quest Diagnostics (UGT1A1 Genotype)	Irinotecan dose adjustment in FOLFIRI and any irinotecan-based HIPEC protocol.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
UGT1A1 genotyping	ARUP Laboratories (UGT1A1 (TA)n Repeat)	Irinotecan dose adjustment in FOLFIRI and any irinotecan-based HIPEC protocol.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-522-2787
Tumor-informed serial ctDNA MRD bracketing CRS-HIPEC	Natera (preferred) (Signatera (tumor-informed ctDNA MRD))	Peri-HIPEC recurrence surveillance; informs adjuvant-therapy duration after cytoreductive surgery + HIPEC.	test info · 13011 McCallen Pass, Building A, Austin, TX 78753 · 1-650-249-9090
Tumor-informed ctDNA MRD	Guardant Health (Guardant Reveal (tumor-naive MRD))	Peri-HIPEC recurrence surveillance; informs adjuvant-therapy duration after cytoreductive surgery + HIPEC.	test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887
Tumor-informed ctDNA MRD	Foundation Medicine (FoundationOne Tracker)	Peri-HIPEC recurrence surveillance; informs adjuvant-therapy duration after cytoreductive surgery + HIPEC.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor-informed ctDNA MRD	Tempus Labs (Tempus xM)	Peri-HIPEC recurrence surveillance; informs adjuvant-therapy duration after cytoreductive surgery + HIPEC.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
CMS transcriptomic classifier (or IHC surrogate panel)	Caris Life Sciences (preferred) (MI Cancer Seek (WES + WTS, CMS classification on whole-transcriptome RNA))	TGF-beta-axis / mesenchymal-targeting trial selection; CMS4 prognostic stratification; CDX2-negative HR-poor subgroup identification.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
CMS classifier	Tempus Labs (Tempus xR (RNA sequencing))	TGF-beta-axis / mesenchymal-targeting trial selection; CMS4 prognostic stratification; CDX2-negative HR-poor subgroup identification.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
CMS classifier	NeoGenomics Laboratories (CDX2 IHC (plus ZEB1, HTR2B, FRMD6, pan-cytokeratin for the IHC-CMS surrogate))	TGF-beta-axis / mesenchymal-targeting trial selection; CMS4 prognostic stratification; CDX2-negative HR-poor subgroup identification.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
SMAD4 (DPC4) IHC	NeoGenomics Laboratories (preferred) (SMAD4 (DPC4) IHC)	Confirms SMAD4-loss phenotype underpinning CMS4 / mesenchymal interpretation and TGF-beta-axis trial discussion.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
SMAD4 IHC	Mayo Clinic Laboratories (SMAD4 IHC)	Confirms SMAD4-loss phenotype underpinning CMS4 / mesenchymal interpretation and TGF-beta-axis trial discussion.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710

SMAD4 IHC	Labcorp (SMAD4 IHC)	Confirms SMAD4-loss phenotype underpinning CMS4 / mesenchymal interpretation and TGF-beta-axis trial discussion.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
20q amplification segment definition (20q11.21 BCL2L1 / TOP1 vs extended 20q13)	Foundation Medicine (preferred) (FoundationOne CDx (copy-number plot))	BCL-xL-directed strategy framing (navitoclax, DT2216) and 20q13-driven AURKA-class trial framing; refines the irinotecan-sensitivity rationale on top of TOP1.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
20q segment definition	Tempus Labs (Tempus xT (segment-level CNV))	BCL-xL-directed strategy framing (navitoclax, DT2216) and 20q13-driven AURKA-class trial framing; refines the irinotecan-sensitivity rationale on top of TOP1.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
20q segment definition	NeoGenomics Laboratories (OncoScan CNV Plus)	BCL-xL-directed strategy framing (navitoclax, DT2216) and 20q13-driven AURKA-class trial framing; refines the irinotecan-sensitivity rationale on top of TOP1.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
High-resolution HLA-A typing (4-digit) plus full class-I / class-II profile	Labcorp (preferred) (HLA A, B, C, DRB1 Profile (High Resolution))	HLA-restricted TCR-T (afami-cel, lete-cel, IMA203) and ImmTAC / soluble-TCR (brenetafusp, IMC-P115C) trial eligibility.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
HLA-A typing	Histogenetics (HLA Class I/II High-Resolution Sanger / NGS Typing)	HLA-restricted TCR-T (afami-cel, lete-cel, IMA203) and ImmTAC / soluble-TCR (brenetafusp, IMC-P115C) trial eligibility.	test info · 300 Executive Boulevard, Ossining, NY 10562 · 1-914-762-1600
HLA-A typing	Discovery Life Sciences (Clinical Trial HLA Typing)	HLA-restricted TCR-T (afami-cel, lete-cel, IMA203) and ImmTAC / soluble-TCR (brenetafusp, IMC-P115C) trial eligibility.	test info · 2904 Beech Court NW, Huntsville, AL 35805 · 1-256-705-4060
Serum CEA, CA 19-9, and CA 125 baseline + serial	Quest Diagnostics (preferred) (CEA / CA 19-9 / CA 125 Serum Panels)	Peri-HIPEC biochemical surveillance; CA 125 specifically tracks peritoneal-disease burden.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Tumor markers	Labcorp (CEA / CA 19-9 / CA 125 Serum Markers)	Peri-HIPEC biochemical surveillance; CA 125 specifically tracks peritoneal-disease burden.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Tumor markers	Mayo Clinic Laboratories (CEA / CA 19-9 / CA 125)	Peri-HIPEC biochemical surveillance; CA 125 specifically tracks peritoneal-disease burden.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
APC E1295 variant resolution + second-allele status	Foundation Medicine (preferred) (FoundationOne CDx)	Wnt-axis investigational trial fit (PORCN, tankyrase, frizzled / WNT974 / LGK974-class).	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639

APC E1295 variant resolution	Tempus Labs (Tempus xT CDx)	Wnt-axis investigational trial fit (PORCN, tankyrase, frizzled / WNT974 / LGK974-class).	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
APC E1295 variant resolution	Caris Life Sciences (MI Cancer Seek)	Wnt-axis investigational trial fit (PORCN, tankyrase, frizzled / WNT974 / LGK974-class).	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
Spatial multiplex immune profile (TIL, CD3/CD8, FAP CAF, pSMAD2/3)	Akoya Biosciences (preferred) (PhenoCycler-Fusion / Phenoptics multiplex IHC)	Anti-TGF-beta + ICI combination trial selection (bintrafusp alfa-class, NIS793, BCA101).	test info · 100 Campus Drive, 6th Floor, Marlborough, MA 01752 · 1-855-896-8401
Spatial multiplex immune profile	NeoGenomics Laboratories (MultiOmyx Multiplex IHC)	Anti-TGF-beta + ICI combination trial selection (bintrafusp alfa-class, NIS793, BCA101).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Hereditary cancer panel POLE / POLD1 proofreading-domain coverage review	Labcorp Genetics (formerly Invitae) (preferred) (Invitae Common Hereditary Cancers Panel (POLE/POLD1 included))	Polymerase-proofreading polyposis surveillance and family cascade decisions; downstream tumor-agnostic ICI rationale if POLE proofreading deficiency is uncovered.	test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037
Hereditary cancer panel review	GeneDx (GeneDx Comprehensive Common Cancer Panel)	Polymerase-proofreading polyposis surveillance and family cascade decisions; downstream tumor-agnostic ICI rationale if POLE proofreading deficiency is uncovered.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 1-888-729-1206
Hereditary cancer panel review	Ambry Genetics (CancerNext-Expanded)	Polymerase-proofreading polyposis surveillance and family cascade decisions; downstream tumor-agnostic ICI rationale if POLE proofreading deficiency is uncovered.	test info · 15 Argonaut, Aliso Viejo, CA 92656 · 1-866-262-7943

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
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Comprehensive tissue NGS (KRAS A59T VAF + extended RAS + BRAF V600 + ERBB2 + PIK3CA + TP53 R273 + SMAD4 + 20q + TMB mut/Mb)	A59 substitutions are recognized RAS-pathway activating alterations and current NCCN consensus treats them as resistance variants for anti-EGFR therapy; a clean VAF and a fully resolved extended RAS plus BRAF V600 are the only way to scope cetuximab / panitumumab and BEACON eligibility. The same panel returns ERBB2 amplification, MSI by NGS, TMB in mut/Mb, and the PIK3CA / APC / TP53 / SMAD4 confirmations the downstream rows depend on.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	1 FFPE block or 15-20 unstained slides; archival acceptable from resected liver met or primary
HER2 (ERBB2) IHC clone 4B5 with reflex dual-probe FISH per HERACLES	HER2 amplification covers another 3-5% of mCRC and is the single most-actionable target check not yet done; a positive return routes trastuzumab deruxtecan via DESTINY-CRC02 (NCT04744831, ORR ~38% including RAS-mutant under the 2024 pan-tumor accelerated approval). MOUNTAINEER tucatinib + trastuzumab is RAS-WT-restricted by label so the A59T patient is closed out regardless of HER2 status. Order IHC first; FISH reflexes if 2+ equivocal.	NeoGenomics Laboratories (HER2 IHC + FISH HERACLES/MOUNTAINEER scoring) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	2-3 unstained slides from archival FFPE

TP53 R273 specific amino acid substitution call	R273H and R273C have the strongest preclinical APR-246 reactivation data; R273L is intermediate; R273S is less-studied. PYNNAACLE rezatapopt is Y220C-restricted and KRAS-WT-only so the patient is structurally ineligible regardless, but the substitution call sharpens any future ATR / WEE1 / HSP90 synthetic-lethal trial discussion at recurrence.	Foundation Medicine (FoundationOne CDx TP53 codon 273 amino acid call) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared with master NGS block
MMR protein IHC four-antibody panel (MLH1 / MSH2 / MSH6 / PMS2)	MSS by a single method misses the rare but real discordant case where NGS-MSI or IHC alone calls a tumor MSS that the orthogonal method calls MSI-H (concordance ~95%, discordance ~5%). Confirming MSS by both methods closes the ICI door definitively rather than presumptively; a discordant MSI-H call reverses the entire decision tree.	NeoGenomics Laboratories (MMR IHC Panel) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	1-2 unstained slides from archival FFPE
TMB explicit value in mutations per megabase with panel and threshold documented	Intake says 'TMB-low' without a number, and 8-9 mut/Mb closes the tumor-agnostic pembrolizumab door while 10-12 opens it. Each panel uses a slightly different bait region and counting rule so the panel name and threshold travel with the number. The FDA label cites FoundationOne CDx specifically for the TMB-H call.	Foundation Medicine (FoundationOne CDx FDA-labeled TMB-H companion diagnostic) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared with master NGS block

PD-L1 IHC with explicit clone (22C3, SP263, or 28-8) and explicit scoring system (CPS for CRC)	'Negative' without a clone or scoring system is not interpretable for trial eligibility; different ICI labels use different cutoffs and antibodies. Several MSS-mCRC combination ICI trials use PD-L1 CPS thresholds for stratification rather than hard exclusion, so the exact CPS value still matters even when the headline is negative. Default to 22C3 with CPS scoring for CRC context.	NeoGenomics Laboratories (PD-L1 22C3 pharmDx CPS + SP263 TPS) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	1-2 unstained slides
DPYD four-variant CPIC genotyping (c.1905+1G>A / 2A, c.1679T>G / 13, c.2846A>T, c.1236G>A / HapB3)	March 2024 FDA fluoropyrimidine label update and 2026 NCCN colon guideline both advise pre-treatment testing. The patient is tolerating FOLFIRI now, but heterozygous c.2846A>T or HapB3 carriers can show progressive cumulative toxicity, and the HIPEC-window dose decision plus any future capecitabine maintenance both benefit from a documented genotype.	Mayo Clinic Laboratories (DPYD Genotype DPYDZ) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	5 mL whole blood (EDTA) or buccal swab
UGT1A1 6 / 28 genotyping	28/28 (and 6/28 in Asian ancestry) confers a 3-4x increased risk of severe irinotecan-related neutropenia and diarrhea; the FDA irinotecan label recommends a starting-dose reduction in homozygotes. Bundle with DPYD on a single pharmacogenomic order. Pre-resolves the irinotecan-HIPEC question if that perfusate route enters discussion.	Mayo Clinic Laboratories (UGT1A1 Genotyping UGTGT) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	5 mL whole blood (EDTA) or shared with DPYD order

20q amplification segment definition (20q11.21 BCL2L1 / TOP1 vs extended 20q13)	20q11.21-restricted amplification centered on BCL2L1 (BCL-xL) and TOP1 is biologically distinct from broad 20q gain that sweeps in 20q13. The BCL2L1 subset supports BH3-mimetic / BCL-xL-degrader rationale; TOP1 amplification correlates with irinotecan sensitivity; the broader 20q13 gain raises AURKA-class trial questions instead. Action is on interpreting the existing CNV plot; reflex to OncoScan only if segment boundaries are ambiguous.	Foundation Medicine (FoundationOne CDx copy-number plot) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared with master NGS block
Tumor-informed serial ctDNA MRD bracketing CRS-HIPEC	Peritoneal-spread CRC is one of the settings where ctDNA MRD is most predictive; pre-CRS detection correlates with shorter PFS and identifies patients who may benefit more from extended systemic therapy than from upfront cytoreduction. Signatera has Medicare coverage in stage II-IV CRC and the tumor-informed bespoke design (16 patient-specific variants from WES) gives the best low-tumor-fraction sensitivity. Bespoke build runs 4-6 weeks; the order has to land BEFORE CRS for the baseline draw to be interpretable.	Natera (Signatera tumor-informed ctDNA MRD) · test info · 13011 McCallen Pass, Building A, Austin, TX 78753 · 1-650-249-9090	5-10 mL plasma per draw; Signatera baseline also needs an FFPE block (can share the master NGS block)

CMS transcriptomic classifier (CRCassigmer / CMScaller) or IHC surrogate panel (CDX2 + ZEB1 + HTR2B + FRMD6 + pan-cytokeratin)	Peritoneal-dominant disease with SMAD4 loss raises the prior for CMS4 (mesenchymal), which identifies patients with a TGF-beta-mediated immune-excluded phenotype and a different combination-therapy landscape than CMS2 / CMS3. The IHC surrogate achieves 87% concordance with transcriptomic CMS when full RNA-seq is unavailable. Best value if Caris MI Cancer Seek is the master panel since CMS comes off the whole-transcriptome data.	Caris Life Sciences (MI Cancer Seek WES + WTS, CMS on whole-transcriptome RNA) · test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669	archival FFPE; 5-10 unstained slides for IHC surrogate
SMAD4 (DPC4) IHC (clone B-8 or EP618Y)	R361H is a known loss-of-function MH2-domain hotspot; SMAD4 IHC loss is the protein-level orthogonal confirmation that the molecular call translates to absent nuclear SMAD4. Cheap defensive call; pair on a single block release with the MMR IHC.	NeoGenomics Laboratories (SMAD4 DPC4 IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	1-2 unstained slides
High-resolution HLA-A typing (4-digit) plus full class-I / class-II profile	MSS / TMB-low mCRC is the population where HLA-restricted TCR-T (afami-cel, lete-cel, IMA203) and ImmTAC (brenetafusp) trials are most often considered; HLA-A*02:01 is the dominant restriction. Bank the result before HIPEC even if no TCR-T trial is in immediate play; the result lasts for life and removes friction from any future referral.	Labcorp (HLA A B C DRB1 Profile High Resolution) · test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167	5-10 mL whole blood (EDTA)

Serum CEA, CA 19-9, and CA 125 baseline + serial	CA 125 elevation correlates with peritoneal carcinomatosis burden and tracks response and recurrence independently of CEA. Together with CA 19-9 they give a low-cost peri-HIPEC response signal that complements ctDNA and imaging. Pull all three at every routine draw rather than CEA alone; the marginal cost is negligible.	Quest Diagnostics (CEA / CA 19-9 / CA 125 Serum Panels) · test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378	3-5 mL serum per marker
APC E1295 variant resolution + second-allele status	Codon 1295 sits within the APC mutation cluster region (codons 1250-1500); truncating events in the MCR retain partial beta-catenin binding but lose the SAMP repeats needed for downregulation, and bi-allelic MCR truncation is the typical 'just right' Wnt-pathway state most studied for Wnt-axis drug rationale. Request the lab to explicitly call the variant type (nonsense / frameshift / missense) and the second-allele state.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared with master NGS block
Hereditary cancer panel POLE / POLD1 coverage review	A 37-year-old with mCRC sits in the early-onset bracket where polymerase-proofreading deficiency (germline POLE / POLD1) is on the differential even at TMB-low. If the prior multi-gene panel did not specifically cover the POLE proofreading exonuclease domain and POLD1, a focused add-on is warranted; if it did, this is a documentation row.	Labcorp Genetics (formerly Invitae) (Invitae Common Hereditary Cancers Panel POLE/POLD1 included) · test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037	no new specimen unless add-on testing needed

TP53 Y220C co-mutation rule-out	Most TP53-mutant CRC tumors carry a single dominant alteration; the intake reports R273 only. PYNACLE rezatapopt requires Y220C plus KRAS wild-type; the case fails the KRAS-WT gate regardless. Cheap to confirm there is no second hit at Y220C from the existing NGS report so the trial screener does not waste time on it.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared with master NGS block
Spatial multiplex immune profile (TIL, CD3 / CD8, FAP CAF, pSMAD2/3)	SMAD4 loss plus a CMS4 phenotype is the canonical TGF-beta-driven immune-excluded state where CD8+ T cells are stuck at the invasive margin behind a FAP+ CAF stromal collar. A multiplex profile separates 'cold' from 'excluded' cases and sharpens any anti-TGF-beta + ICI combination-trial discussion. Research-grade; reserve for the moment a TGF-beta-axis protocol enters serious discussion.	Akoya Biosciences (PhenoCycler-Fusion multiplex IHC) · test info · 100 Campus Drive, 6th Floor, Marlborough, MA 01752 · 1-855-896-8401	1 FFPE block or 5-10 unstained slides

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.